

Dear Editors,

Please consider my manuscript, “Protective Computing: An Architectural Framework for Safety-Critical Systems Under Conditions of Human Vulnerability,” for publication as a Research Article.

This paper introduces Protective Computing as a systems-engineering orientation designed for contexts in which human–computer interaction occurs under vulnerability (medical crisis, coercion, infrastructural instability, or cognitive degradation). The manuscript formalizes Stability Bias ($H = f(D, V, I)$), defines a Vulnerability State Machine (S_0 – S_3), specifies five normative design principles using RFC-style language, and proposes a provisional evaluation metric (Protective Legitimacy Score).

The paper synthesizes and extends work in safety-critical HCI, resilience engineering, local-first architecture, privacy-by-design, and digital safety research. It advances the claim that architectural legitimacy in safety-critical domains must be evaluated by system behavior under degraded human and infrastructural conditions rather than ideal stability assumptions.

A reference implementation (PainTracker) is provided to demonstrate technical feasibility. The implementation evidence is presented conservatively and is not claimed as large-scale validation.

The manuscript has not been published elsewhere and is not under consideration by another journal. There are no conflicts of interest to declare.

I believe this work will be of interest to researchers in HCI, trustworthy systems, digital medicine, and safety-critical computing.

Thank you for your consideration.

Sincerely,
K. Overton